Data Visualization Assignment

Goodreads Spoiler Reviews Dataset – UC San Diego (McAuley Lab)

Dataset link: https://mcauleylab.ucsd.edu/public_datasets/gdrive/goodreads/goodreads_reviews_spoiler.json.gz

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Section A: Data Understanding

1. Describe the schema of a single review record in the Goodreads spoiler dataset. In what format does the `review_sentences` field store spoiler information?

2. What is the difference between the `has_spoiler` field (per-review boolean) and the individual spoiler flags (0/1) inside `review_sentences` (per-sentence)? Based on how the two fields are related, would you expect `has_spoiler=True` whenever at least one sentence is flagged 1?

3. The real dataset contains ˜1.38 million reviews across ˜25,000 books and ˜19,000 users, totaling ˜17.6 million sentences (source: Wan et al., ACL'19). Why would loading a file this large directly into memory as a Python list be risky on a typical laptop? What strategy did we use to load it safely (see the `MAX_ROWS` variable)?

Section B: Feature Engineering

4. How are the `review_length` and `spoiler_ratio` features computed? Write down the formula for each.

5. If a review has 0 sentences, why does the `spoiler_ratio` function not crash? Which safeguard in the code prevents this?

Section C: Distribution Plots (Histogram & Boxplot)

6. Looking at the rating histogram, is the distribution of ratings normal or skewed? Which rating value has the highest frequency?

7. Compare the median review length between `has_spoiler=True` and `has_spoiler=False` groups in the boxplot. Which group has longer reviews, and why might that be the case?

8. What do the individual dots (outliers) in the boxplot represent?

Section D: Scatter Plot

9. Does the scatter plot of review length vs. rating show a clear linear relationship? Justify your answer.

10. What additional insight does the color encoding (`has_spoiler`) provide that could not be seen from the x–y position alone?

Section E: Correlation Heatmap

11. Among `rating`, `n_sentences`, `review_length`, and `spoiler_ratio`, which pair shows the strongest positive correlation?

12. Why does the correlation coefficient always lie between -1 and $+1$? What does a value close to 0 indicate?

Section F: Leaderboard (Bar Chart)

13. Why was the top-10 most-reviewed-books chart drawn as a horizontal bar chart (`barh`) instead of a vertical one?

14. Is the book with the most reviews necessarily the "best" book? What additional metric should be considered (hint: rating)?

Section G: Radar Chart

15. Why were the metrics normalized to a 0–1 range before plotting on the radar chart, instead of using their original scales (e.g. review count 50–500 vs. average rating 0–5)?

16. How does the shape of a book's radar polygon indicate whether it is "balanced" across metrics or extreme in just one?

Section H: t-SNE Plot

17. What does the t-SNE algorithm do – how does it convert high-dimensional TF-IDF vectors (300 dimensions) into a 2D plot?

18. In this notebook's plot, spoiler and non-spoiler reviews appear as fairly separate clusters. Given the note on data above, is this separation likely to be as clean on the real dataset (where spoilers are only ˜3.22% of sentences and wording overlaps more)? Explain your reasoning.

19. What role does the `perplexity=30` parameter play in t-SNE? How might changing it affect the resulting plot?

Section I: Analysis & Critical Thinking

20. If you were a publisher trying to determine which book is most "spoiler-prone," which two visualizations from this assignment would be most useful, and why?

21. This assignment uses synthetic sample data with the same schema as the real dataset. How might your conclusions change if you re-ran the notebook on the real UCSD dataset?